

A Systematic Framework for Design of Experiments in System Design: Establishing Order and Relevance in First-Order Analysis

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ABSTRACT

The rising complexity of modern weapons systems necessitates robust, data-driven methods to efficiently explore and optimize design parameters. We present a systematic framework that integrates Design of Experiments (DoE) with Bayesian network modeling to identify critical interactions and refine system performance. Our approach demonstrates how factor screening and conditional probability analysis can reduce experimental overhead while maintaining statistical fidelity. In contrast to conventional one-factor-at-a-time strategies, this integrated technique leverages structured sampling and probabilistic inference, substantially improving analytical rigor and scalability. Our results confirm that the proposed methodology offers a reliable and cost-effective path to designing advanced weapon systems, thereby enhancing operational readiness and decision-making.

Keywords: Design of Experiments, System Design, First-Order Analysis, Modeling and Simulation, Aircraft Survivability

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1. INTRODUCTION

Modern weapons systems have grown in both complexity and performance requirements, prompting the need for robust, data-driven methodologies to efficiently explore and optimize the design space. This paper presents a systematic framework that combines *Design of Experiments (DoE)* with Bayesian network modeling to uncover critical interactions and refine system-level performance. By focusing on factor screening, conditional probability analysis, and structured sampling, the proposed approach demonstrates how first-order analysis can effectively reduce experimental overhead while maintaining statistical fidelity.¹⁻⁴

Recent advancements in computational modeling and simulation have enabled large-scale experimentation, offering a deeper understanding of key drivers such as propulsion parameters, detection thresholds, and engagement dynamics. However, traditional one-factor-at-a-time (OFAT) methods are limited by their inability to account for factor interactions and by the significant number of runs needed for higher-dimensional trade spaces. By contrast, DoE systematically investigates multiple factors and their interactions in a single coherent study. This structured approach not only yields quantitative insights, such as main and interaction effects, but also guides the identification of dominant variables under constrained budgets or computational resources.

Within this framework, Bayesian networks augment DoE by representing conditional probabilities across sequential phases of an engagement. These probabilistic models track outcomes such as weapon activation, detection, launch, intercept, and eventual kill or survival. When integrated with DoE, Bayesian networks enable analysts to couple factor screening (e.g., altitude, geometry, sensor placement) with probabilistic assessments, facilitating a data-driven prioritization of design parameters. The result is a more holistic evaluation of how design choices and operational tactics converge to shape system effectiveness.

The remainder of the paper is structured as follows. Section 2 provides a detailed overview of the DoE principles, including problem definition, factor selection, and experimental design strategies. It then introduces the one-on-one Bayesian network scenario used to demonstrate how conditional probabilities inform system survivability. Section 3 describes the methodology employed in this study, from specifying the key hypotheses and selecting factors, levels, and ranges, to outlining the simulation execution plan and data collection scheme.

Section 4 presents the results of the analysis, highlighting detection outcomes, feature importances, and the interplay among critical variables. Finally, Section 5 discusses broader implications of these findings, offering conclusions and avenues for future work related to aircraft survivability, defense system design, and advanced probabilistic modeling of complex multi-factor systems.

2. BACKGROUND

This section provides a comprehensive overview of the fundamental principles and technical methodologies that underpin the DoE framework. The discussion is intended as an end-to-end guide to DoE, integrating theoretical principles with practical implementation considerations. The section begins with a general introduction to DoE and its critical role in systematic experimentation, followed by detailed discussions on the essential components of the framework: Problem Definition, Factors, Levels, and Ranges, Experimental Design, Execution Plan, Data Collection, and Data Analysis. This section concludes with the relevant background information for a one-on-one engagement scenario that will be used to demonstrate the implementation of DoE in the context of weapon optimization.

2.1 Design of Experiments

Design of Experiments (DoE) is a structured methodology used to investigate the relationships between multiple input variables and one or more response variables. The approach was initially developed by Fisher, who introduced the principles of *randomization*, *replication*, *blocking*, and *factorial experiments* as a systematic means of studying multiple factors simultaneously.¹ Building on Fisher’s foundational work, Box, Hunter, and Hunter formalized the practice of *sequential experimentation*, which iteratively refines models as new data are collected, and they popularized *fractional factorials* for efficiently screening high-dimensional factor spaces while preserving critical effects.^{2,3} Their advocacy of *response surface methodology* (RSM) further enabled researchers to optimize continuous factors by approximating system responses with polynomial functions, and they emphasized *robust design* to manage noise factors, ensuring reliable performance under varying operating conditions.

The primary purpose of DoE is to *maximize the information gained* from experimental resources by systematically isolating, measuring, and interpreting the influence of different factors. This approach not only identifies which factors have the most significant effects on the response variable, but also uncovers how these factors *interact* with each other, providing deeper insight into the behavior of complex systems. To achieve these objectives, DoE employs mathematical frameworks that guide the design and interpretation of experiments.

One such framework is the *full factorial design*, which systematically evaluates all possible combinations of factor levels. If each of the k factors is examined at L_i distinct levels, the total number of experimental runs is:

$$N = \prod_{i=1}^k L_i, \quad (1)$$

where N grows exponentially with the number of factors and their respective levels. Although potentially expensive in terms of experimental runs, this exhaustive approach reveals both *main effects* (each factor’s direct influence on the response) and *interaction effects* (how two or more factors together influence the response).

Further insight into these main and interaction effects is gained through the generalized linear model:

$$Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \sum_{i < j} \beta_{ij} X_i X_j + \cdots + \epsilon, \quad (2)$$

where Y denotes the response variable, β_0 is the intercept, β_i represent the main-effect coefficients, and β_{ij} capture the interactions between different factors X_i and X_j . The error term ϵ is assumed to follow a normal distribution with mean zero and constant variance. This model framework enables a clear, quantitative dissection of how each factor and combination of factors affects the outcome, thereby guiding optimization strategies.

These principles form the foundation of modern experimental design, which guides researchers in systematically evaluating factor influences while improving the efficiency and reliability of empirical studies.^{3,4} In

industrial experimentation, engineering, and scientific research, the adoption of factorial designs, fractional factorials, and RSM has transformed how complex systems are optimized and how decisions are made about process improvements and resource allocation. By uncovering the most influential factors and interactions, DoE offers a powerful data-driven basis to improve system performance and ensure repeatable and robust results.

2.2 Problem Definition

The Problem Definition phase is a critical step that establishes the scope and objectives of an experimental study. This phase involves clearly articulating the research questions, specifying the response variable(s), and formulating the hypotheses to be tested. A precise problem definition not only guides the selection of factors and levels, but also ensures that the resulting statistical models are aligned with the underlying research objectives. It forms the basis for constructing a robust and interpretable experimental framework.

2.3 Factors, Levels, and Ranges

A fundamental component of DoE is the systematic identification and quantification of factors, which are independent variables manipulated during experimentation. The foundational principles of experimental factor selection and control were first established by Fisher,¹ later refined and expanded by Box, Hunter, and Hunter,^{2,3} who introduced practical approaches to factor screening, optimization, and response surface methodologies.

Each factor is characterized by its range of values, which may be continuous or discrete. For a continuous factor X , its range is defined from a minimum value $X_{\min}$ to a maximum value $X_{\max}$, with intermediate levels selected based on experimental requirements or theoretical considerations. The proper selection of factor levels is crucial, as Box, Hunter and Hunter emphasized the importance of ensuring that the experimental trade space remains both comprehensive and representative of all relevant conditions.²

A well-structured factorial design allows for the systematic exploration of factor effects and interactions, enhancing the efficiency and accuracy of the experiment. The selection of factor levels directly influences the resolution of the experimental model and its ability to detect significant effects, reinforcing the necessity of statistically rigorous methodologies in experimental design.³

2.4 Experimental Design

Experimental Design provides the strategic blueprint for organizing and conducting experiments. The foundational principles of experimental design were first introduced by Fisher,¹ who established the framework for factorial experiments, blocking, and randomization. Later, Box, Hunter, and Hunter^{2,3} expanded these methodologies, refining the application of factorial and fractional factorial designs for industrial and scientific experimentation.

Various design strategies exist, each offering distinct advantages depending on the objectives and constraints of the experiment. In a full factorial design, all possible combinations of factor levels are explored, providing a complete interaction analysis at the cost of a potentially large number of experimental runs. Fractional factorial designs mitigate this issue by selectively confounding higher-order interactions, significantly reducing the number of trials while retaining the ability to estimate primary and lower-order effects.³

Alternatively, a randomized block design partitions the experimental runs into blocks, where each block is homogeneous with respect to extraneous variables. This technique, first introduced by Fisher,¹ is particularly effective in reducing variability and enhancing the precision of effect estimates by controlling nuisance factors.

The choice of design represents a trade-off between experimental cost and the desired level of statistical fidelity. When computational or resource constraints are present, carefully selecting an appropriate design, such as a fractional factorial or randomized block design, ensures that meaningful insights are obtained without unnecessary experimental overhead.²

2.5 Execution Plan

The Execution Plan outlines the operational details necessary to implement the experimental design in practice. This plan includes the development of experimental or simulation protocols, the allocation of computational or laboratory resources, and the scheduling of experimental runs. In many contemporary applications, computational platforms such as Python are used to implement mathematical models and algorithms that simulate the experimental conditions. The execution plan ensures that the experimental procedure is systematic, reproducible, and seamlessly integrated into the broader research framework.

2.6 Data Collection

Data Collection is the systematic process of capturing the outputs of each experimental run. This process involves recording the values of the input factors along with the corresponding response measurements. Automated data collection, often implemented through software, minimizes human error and ensures that the dataset is comprehensive and structured. The quality and integrity of the collected data are critical, as they directly influence the accuracy of subsequent statistical analyses and model validations.

2.7 Data Analysis

Data Analysis encompasses a suite of statistical and computational techniques used to interpret experimental data and extract actionable insights. The foundations of statistical data analysis were laid by Fisher,¹ who introduced analysis of variance (ANOVA) and regression techniques to quantify experimental effects systematically. Later, Box, Hunter, and Hunter^{2,3} expanded these methodologies, incorporating statistical modeling, residual analysis, and the development of response surfaces for complex experimental data.

Typically, initial data analysis involves computing summary statistics such as means, variances, and percentiles to assess the distribution and range of the collected data. A widely used statistical model in DoE is the linear regression model, formulated as:

$$Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \epsilon, \quad (3)$$

where Y represents the response variable, β_0 is the intercept, β_i are regression coefficients associated with input factors X_i , and ϵ is the error term, which is assumed to follow a normal distribution with mean zero and constant variance.³

Beyond traditional regression modeling, advanced techniques such as the development of metamodels, including polynomial response surfaces and kriging models, are used to approximate the relationship between input factors and the response variable, facilitating rapid interpolation and prediction for new experimental conditions.²

Ensuring the validity of the model assumptions is essential for robust statistical inference. Assumptions such as normality, independence, and homoscedasticity must be rigorously validated through residual analysis, hypothesis testing, and graphical diagnostic tools. Failure to meet these assumptions can lead to biased estimates and misleading conclusions, underscoring the importance of proper model diagnostics in experimental analysis.³

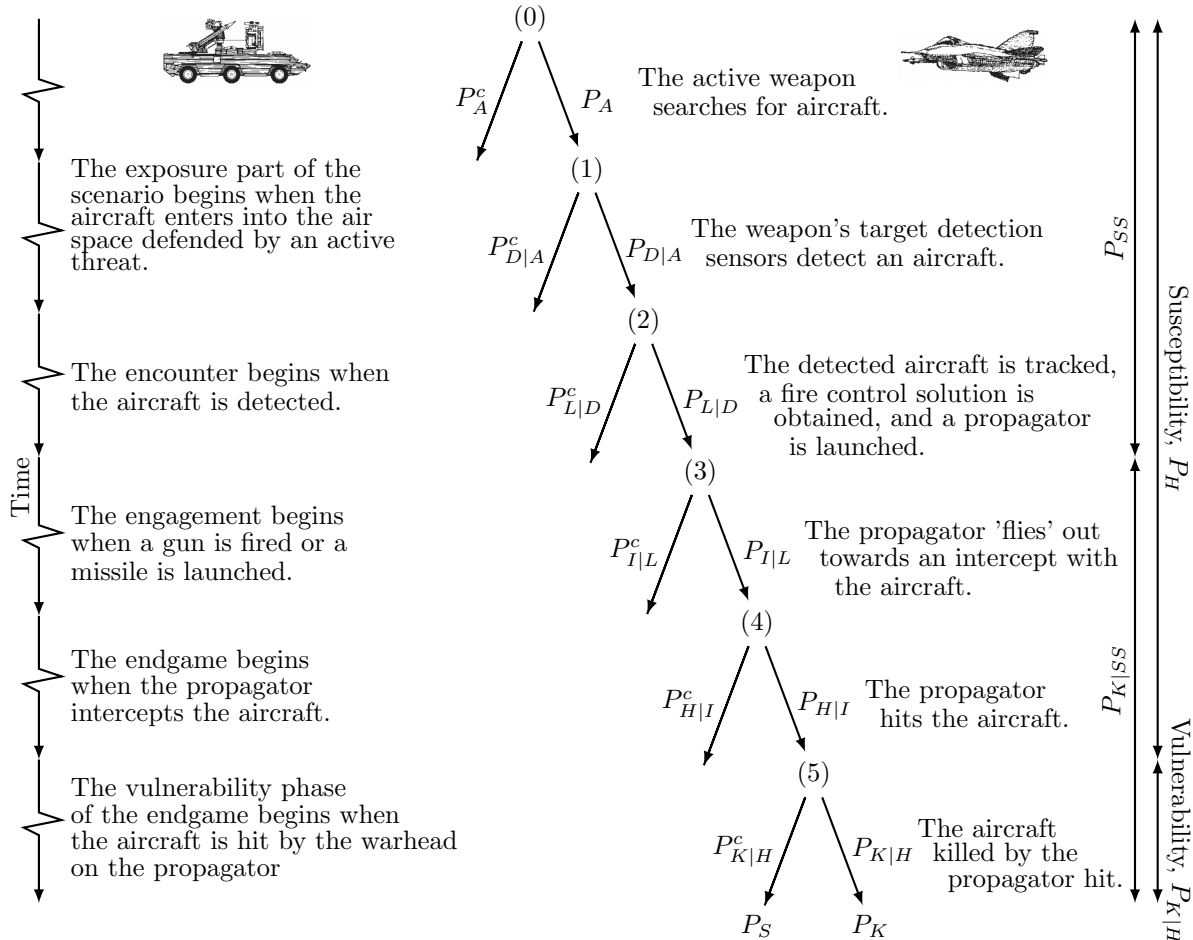
2.8 The One-on-One Bayesian Network Scenario

Figure 1 illustrates the one-on-one Bayesian network scenario, representing the sequential decision-making process in an air defense engagement. This probabilistic tree model defines the six critical phases of an encounter between an aircraft and an enemy weapon system, where each phase has two possible outcomes: a *successful event* (right branch) or an *unsuccessful event* (left branch).

For instance, the aircraft is exposed to an active, searching threat with probability P_A . Here, *exposure* refers to the condition where the aircraft enters the defended area of an active threat. If the aircraft is not exposed to an active threat, this occurs with probability P_A^c , where:

$$P_A^c = 1 - P_A. \quad (4)$$

Each branch in the tree diagram represents a conditional probability event, determining whether the aircraft advances through the engagement phases or avoids destruction. The probabilities assigned to each branch quantify the likelihood of success or failure at a given stage. From the perspective of the air defense system, a successful engagement occurs only if all six phases result in a successful outcome, ultimately leading to the aircraft being destroyed. Conversely, failure at any phase ensures the aircraft's survival.



where $P_S = P_K^c = 1 - P_K$ (The aircraft survives the scenario.) and
 $P_K = P_A P_{D|A} P_{L|D} P_{I|L} P_{H|I} P_{K|H}$ (The aircraft is killed in the scenario.)

Figure 1. Tree diagram for the one-on-one scenario (single shot) - Original figure from Ball (2003).⁵

2.8.1 Phase Probabilities in the Engagement Process

Each phase in the one-on-one scenario represents a distinct event in the engagement process, modeled using conditional probabilities that define the likelihood of an aircraft progressing through each stage:

1. **Weapon Activation Probability (P_A):** The probability that a threat weapon is active and searching for aircraft within its defended airspace.
2. **Detection Probability ($P_{D|A}$):** Given that the weapon is active, the probability that the aircraft is detected by the threat system.
3. **Launch Probability ($P_{L|D}$):** Given that the aircraft is detected, the probability that the weapon system tracks the aircraft, computes a fire control solution, and launches a propagator (e.g., missile or gunfire).

4. **Intercept Probability** ($P_{I|L}$): Given that a propagator was launched, the probability that it reaches the aircraft's flight path.
5. **Hit Probability** ($P_{H|I}$): Given that the propagator intercepts the aircraft, the probability that the aircraft is hit directly or affected by a proximity warhead detonation.
6. **Kill Probability** ($P_{K|H}$): Given that the aircraft is hit, the probability that the hit destroys the aircraft.

The probability that the aircraft is *hit at least once* as it traverses the defended area is:

$$P_H = P_A P_{D|A} P_{L|D} P_{I|L} P_{H|I}. \quad (5)$$

Once a hit occurs, the probability of *killing* the aircraft is given by $P_{K|H}$. Combining the probability of a hit P_H with the conditional kill probability $P_{K|H}$ yields the overall likelihood of the aircraft being destroyed:

$$P_K = P_H \times P_{K|H} = P_A P_{D|A} P_{L|D} P_{I|L} P_{H|I} P_{K|H}. \quad (6)$$

Conversely, the probability of survival is defined as:

$$P_S = P_K^c = 1 - P_K. \quad (7)$$

2.8.2 Interpreting the Bayesian Network Tree

Each rightward branch in the Bayesian network represents a successful event, indicating that the aircraft advances to the next phase of the engagement. For example, the aircraft may be detected, tracked, intercepted, or hit by the propagator. Conversely, each leftward branch represents an unsuccessful event, where the weapon system fails to detect, track, or hit the aircraft, resulting in the aircraft's survival.

At each phase of the engagement sequence, the sum of the probabilities for success and failure is equal to unity, mathematically expressed as:

$$P + P^c = 1. \quad (8)$$

This ensures that all possible outcomes are accounted for within the probabilistic model. The final outcome of the scenario depends on whether all six engagement phases are successfully completed. If any phase results in failure, the aircraft remains operational; otherwise, it is neutralized.

This Bayesian network model effectively captures the probabilistic nature of air defense engagements, providing analysts with a framework to estimate mission success or failure under various conditions. By evaluating the conditional probabilities at each phase, decision-makers can identify vulnerabilities, optimize countermeasures, and refine operational strategies to enhance aircraft survivability.

3. METHODS

This section presents a structured workflow for determining the probability of detection, $P_{D|A}$, under the simplifying assumption that detection occurs whenever a direct line-of-sight (LoS) exists between an aircraft and a ground observer. By integrating principles from DoE,¹⁻⁴ this methodology details each stage of the experimental process, from the initial problem definition through simulation execution and data analysis, ensuring transparency, reproducibility, and a solid foundation for extending the model's complexity in future research.

3.1 Problem Definition

3.1.1 Objective of the Experiment

The central objective is to determine the *critical separation distance* at which LoS is assured, thereby guaranteeing detection. Concretely, this translates to finding the threshold where a ground observer has an unobstructed view of the aircraft, expressed as $P_{D|A}$. Establishing this threshold is vital for assessing system survivability and operational readiness in defense scenarios.

3.1.2 Hypothesis

We hypothesize that increasing aircraft altitude substantially raises the probability of detection, whereas reduced horizontal separation and optimal headings magnify the likelihood of LoS. Conversely, flying at lower altitudes or at headings unfavorable to the observer’s position diminishes the chance of detection. Thus, altitude, horizontal distance, and heading collectively shape the binary detection outcome.

3.1.3 Success Criteria

The experiment’s success hinges on whether the simulated probability of detection $P_{D|A}$ converges with theoretical line-of-sight expectations. Close alignment between model predictions and classical geometric constraints on LoS confirms that critical system parameters, such as altitude and observer location, are being accurately captured. Achieving this validates the first-order approach and lays the groundwork for refining higher-fidelity models.

3.2 Factors, Levels, and Ranges

To capture the essential contributors to line-of-sight detection, the simulation framework includes several key *factors*, each examined within specified *ranges* and *levels*. Table 1 identifies these variables and their respective domains. Factors such as *Aircraft Altitude*, *Horizontal Separation*, and *Aircraft Heading* are varied in controlled intervals or randomly sampled, while the *Ground Observer Location* is randomly assigned within geographic boundaries.

Table 1. Key Factors (Independent Variables)

Factor	Description	Range/Values
Aircraft Altitude	Vertical position of the aircraft above ground level.	45.72 m (150 ft) to 19,812 m (65,000 ft)
Horizontal Separation	Relative horizontal distance between the aircraft and the observer.	Variable; determined by simulation geometry
Aircraft Heading	Direction of aircraft motion relative to the observer’s location.	Random, 0° to 360°
Ground Observer Location	Geographic coordinates within a defined region representing the observer.	Longitude: [-82, -80], Latitude: [27.5, 29.0]

Control Factors. As shown in Table 2, several variables remain fixed throughout each simulation run to isolate the effects of the primary factors. These include *Observer Altitude*, *Simulation Time Step*, *Total Simulation Duration*, and *Aircraft Speed*. Holding these parameters constant underscores how varying the core factors in Table 1 specifically influences detection.

Table 2. Control Factors

Factor	Description	Fixed Value/Range
Observer Altitude	Height of the ground observer.	1.5 m
Simulation Time Step	Time increment for each simulation iteration.	10 s
Total Simulation Duration	Overall time span for a simulation run.	3600 s (1 hour)
Aircraft Speed	Constant speed of the aircraft during simulation.	250 m/s

Noise Factors. Certain variables remain uncontrolled yet may influence detection outcomes. Table 3 lists such *noise factors*, including geographic variability and environmental conditions, which can be addressed by robust design strategies if future studies require greater fidelity.³

Table 3. Noise Factors

Factor		Description	Considerations
Geographic Variability		Natural variability in the relative position between observer and aircraft.	Incorporate detailed terrain or urban structure effects
Environmental Conditions	Condi-	External factors such as weather, atmospheric clarity, and lighting.	Add variables for cloud cover, fog, or time-of-day effects
Measurement Uncertainty	Uncer-	Inherent errors in recording positions and altitudes.	Introduce sensor error margins or calibration data

3.3 Experimental Design

3.3.1 Type of Design

The experimental process is implemented as a *randomized sample simulation*, building on DoE practices while adapting them to large-scale computational studies.⁴ Initial consideration of full or fractional factorial designs confirmed the potential for exponential growth in runs. Hence, we opted for a randomized design to efficiently capture the system’s variability within feasible computation times. Future enhancements can include blocking or specialized sampling if evidence of systematic bias emerges.

3.3.2 Number of Experimental Runs

We executed 10,000 unique simulation runs to ensure a statistically significant representation of the defined factor space. This choice balances computational resources with the need for robust sampling of altitude, heading, and horizontal separation conditions.

3.3.3 Randomization Plan

To mitigate sampling bias, each run was assigned a unique, pseudo-random seed controlling factor values and observer positions. Randomizing not only the input parameters but also the order of runs increases confidence in the reproducibility and objectivity of the results.¹

3.3.4 Replicates and Blocking

Replicates were deemed unnecessary for this first-order analysis, as the 10,000 runs already provide extensive coverage of factor interactions. Blocking was not adopted because randomization sufficiently distributes any nuisance variability across the parameter space. Subsequent iterations may incorporate these techniques if additional refinement or bias control becomes necessary.²

3.4 Execution Plan

3.4.1 Experimental Setup

The simulation framework is coded in Python using `pandas` for data handling, `matplotlib` for visualization, and `cartopy` for geographic calculations.^{6,7} Each run initializes an aircraft and ground observer position, computes LoS via the haversine formula, and updates positions deterministically based on heading and speed at each time step.

3.4.2 Resources and Environmental Controls

Simulations were performed on standard CPUs; however, the framework is amenable to parallelization if future research demands higher throughput. Environmental parameters, such as simulation time step and fixed aircraft speed, were maintained across all runs to isolate the effects of primary factors and minimize extraneous sources of variation.

3.4.3 Real-Time Monitoring and Contingency Plans

Pre-execution code validation and unit tests minimized runtime errors. No real-time monitoring was required, though a simple contingency plan (e.g., rerun failed jobs) remains feasible if large-scale simulations encounter unusual system load or computational anomalies.⁴

3.5 Data Collection

Each experimental run automatically logs input factors and the resulting detection outcome in a structured `pandas` `DataFrame`. Table 4 summarizes the recorded variables, including positions of the aircraft and ground observer, the aircraft’s altitude, speed, heading, and the binary LoS indicator. This automated logging significantly reduces manual errors, ensuring consistency across all 10,000 runs.

Table 4. Summary of Data Elements Captured in Each Simulation Run

Variable	Description
Ground Latitude	Latitude of the randomly selected ground object location.
Ground Longitude	Longitude of the ground object.
Ground Altitude	Altitude of the ground observer (fixed at 1.5 m).
Aircraft Initial Latitude	Latitude of the aircraft’s starting position within the bounding box.
Aircraft Initial Longitude	Longitude of the aircraft’s starting position within the bounding box.
Aircraft Altitude	Varied between 45.72 m (150 ft) and 19,812 m (65,000 ft).
Aircraft Speed	Held constant at 250 m/s.
Aircraft Heading	Randomized from 0° to 360°.
Detection Outcome	Binary LoS indicator: 0 indicates detection, 1 indicates non-detection.

3.6 Data Analysis

3.6.1 Software and Statistical Methods

We employed Python-based libraries, `numpy`, `pandas`, `matplotlib`, `statsmodels`, and `scikit-learn`, for statistical modeling and visualization.^{6–8} Our approach combined *predictive* methods (Random Forest for feature importance and classification) with *inferential* techniques (OLS regression, ANOVA, and logistic regression) to quantify and interpret the relationships between factors and the detection outcome $P_{D|A}$.

3.6.2 Modeling Approach and Assumption Checks

Models were evaluated under the first-order assumption that a direct LoS guarantees detection. Random Forest classification revealed which factors most strongly drive detection, while regression-based analyses (OLS, logistic) offered statistical significance testing and effect estimates.⁴ Residual plots checked for normality and homoscedasticity in OLS fits, and logistic regression was validated for independence and linearity in the logit. This diagnostic process helps ensure reliable inferences, highlighting whether higher-order terms or blocking might become relevant in future investigations.

3.6.3 Interpretation of Results

By combining these data-driven insights with geometric intuition, we identified how altitude and horizontal separation dominate LoS probability. The results validate the feasibility of using line-of-sight principles as a first approximation for detection. Moreover, the robust sampling design and subsequent analysis confirm the simulation’s capacity to capture key interactions, guiding potential refinements, such as extended fidelity to account for atmospheric scattering, complex terrain, or sensor performance, in future studies.

4. RESULTS AND ANALYSIS

This section presents the findings obtained from the simulation and subsequent statistical evaluations. We begin by outlining the key observations from the confusion matrices and the associated accuracy metrics, then examine the relative importance of the predictors, and finally discuss the implications of these results in terms of optimization and practical relevance.

4.1 Interpretation of Results

4.1.1 Key Findings

A Random Forest classifier was employed to predict whether LoS was established, and thus whether detection occurred. The training set yielded a perfect accuracy of 1.0000, indicating that the model fully captured the patterns present in the training data. However, on the test set, the accuracy was 0.9170, suggesting that while the model performs well, a small subset of boundary or edge cases remains imperfectly captured. Figures 2 and 3 display the confusion matrices for both the training and test sets side by side, illustrating the distribution of correct and incorrect classifications.

		Predicted label	
		Pred=0	Pred=1
True label	True=0	4847	0
	True=1	0	3153

Figure 2. Confusion Matrix for the training set (accuracy = 1.0000).

		Predicted label	
		Pred=0	Pred=1
True label	True=0	1150	80
	True=1	86	684

Figure 3. Confusion Matrix for the test set (accuracy = 0.9170).

To further investigate the role of each predictor, an OLS model was fitted and subjected to ANOVA (Table 5), and a logistic regression model was also evaluated. These analyses consistently identified aircraft altitude as the primary driver of the detection outcome. Other variables, such as ground latitude, ground longitude, initial aircraft position, and heading, exhibited relatively modest effects. The strong significance of aircraft altitude indicates that the geometry of the scenario, particularly in the vertical dimension, dominates the probability of detection, $P_{D|A}$.

Table 5. ANOVA Results for Key Predictors in the OLS Model

Variable	Sum of Squares	F-value	p-value
Ground Latitude	0.1142	0.6099	0.4349
Ground Longitude	0.1366	0.7295	0.3931
Ground Altitude	0.3321	1.7734	0.1830
Aircraft Initial Latitude	0.0730	0.3900	0.5323
Aircraft Initial Longitude	0.1471	0.7857	0.3754
Aircraft Altitude	412.0531	2200.3590	< 0.001
Aircraft Speed	0.3321	1.7734	0.1830
Aircraft Heading	0.4903	2.6182	0.1057

An examination of feature importances from the Random Forest model corroborates these observations. Table 6 ranks the features in descending order of their importance, with aircraft altitude again emerging as the most influential.

4.1.2 Optimization Insights

Given the dominant influence of aircraft altitude, adjusting this parameter emerges as a primary strategy for altering detection probability. In operational terms, an aircraft seeking to avoid detection might benefit from a lower altitude profile when feasible, although the benefits may be limited by additional constraints such as terrain and airspace regulations. Conversely, defense systems can prioritize detection capabilities by optimizing sensor

Table 6. Feature Importances from the Random Forest Classifier

Feature	Importance Score
Aircraft Altitude	0.2972
Aircraft Initial Longitude	0.2396
Aircraft Initial Latitude	0.1578
Aircraft Heading	0.1462
Ground Longitude	0.0820
Ground Latitude	0.0772
Aircraft Speed	0.0000
Ground Altitude	0.0000

placements or coverage angles, particularly in relation to higher altitudes. Moreover, the secondary importance of the initial plane longitude and latitude suggests that these horizontal positioning factors, while not as impactful as altitude, still warrant consideration when refining operational strategies.

4.1.3 Practical vs. Statistical Significance

Although altitude clearly has a highly significant statistical effect, its practical impact depends on real-world constraints such as flight corridors and mission objectives. The difference between, for example, a mid-range altitude and a maximum operational ceiling may be statistically meaningful yet logistically difficult to achieve. The same logic applies to other variables: even if they show weaker significance in the model, their practical relevance may increase under certain tactical or geographic conditions. Thus, while the findings guide an overall strategy for optimizing detection (or avoidance) of an aircraft, further operational analysis is essential to translate these statistically significant outcomes into actionable practices.

4.1.4 Classification Boundary

Figure 4 illustrates the model’s classification boundary when varying the two most influential features identified by the Random Forest: aircraft altitude (horizontal axis) and initial aircraft longitude (vertical axis). Each point on the plot represents a synthetic combination of altitude and longitude, with color indicating the model’s predicted class. The abrupt transitions between the red and blue regions highlight where the classifier changes from predicting one class to the other. As expected, altitude exerts a strong influence on the outcome, particularly at higher values, where the model consistently predicts one class. Longitude also plays a role, though its impact is less pronounced than altitude’s. This also highlights that with decreased altitude the longitudinal variable becomes smaller and smaller, in line with the initial hypothesis.

5. CONCLUSION

This study presented a comprehensive framework for employing Design of Experiments (DoE) in the context of aircraft survivability and system optimization. By systematically defining factors, levels, and ranges, and then applying appropriate experimental designs such as full factorial or randomized block approaches, the methodology demonstrated how key performance indicators can be measured and interpreted with statistical rigor.¹⁻³ The integration of a Bayesian network model further illustrated how conditional probabilities in a sequential engagement (from exposure to kill phases) offer valuable insights into mission success and aircraft survivability.⁵

Central to the approach was the use of standardized processes for problem definition, data collection, and analysis, ensuring that the resulting models are both reproducible and adaptable to varying experimental objectives. The paper highlighted the importance of validating model assumptions to preserve the reliability of statistical inferences. Moreover, advanced techniques like metamodeling and response surface methodologies serve as promising avenues for approximating complex relationships between input variables and responses, thus paving the way for efficient exploration of larger and more intricate trade spaces.

The results underscore the effectiveness of the DoE framework in systematically uncovering critical factors and interactions that affect system-level performance. By coupling this framework with rigorous probability modeling via Bayesian networks, decision-makers can more confidently prioritize resources, optimize countermeasures, and

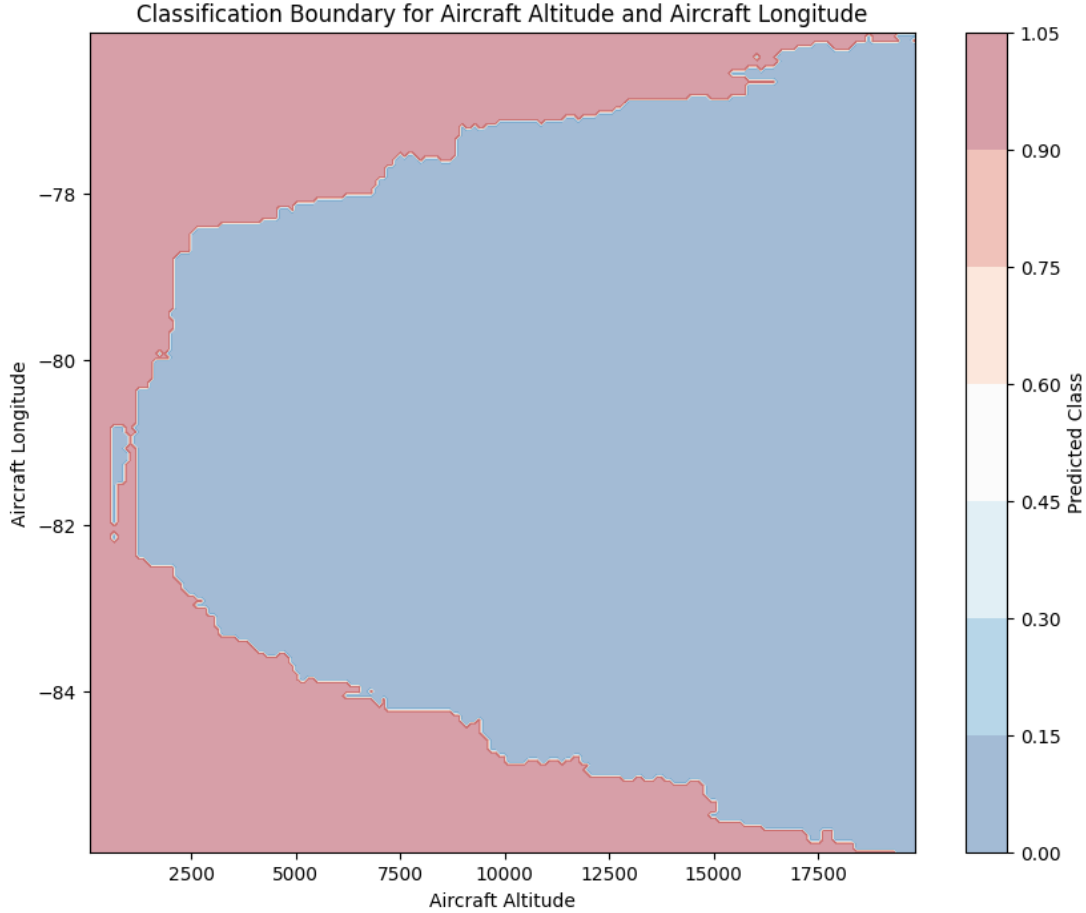


Figure 4. Classification Boundary for Aircraft Altitude and Aircraft Longitude, where red represents no LOS, and blue represents LOS.

refine operational strategies. In doing so, this work offers a scalable, data-driven roadmap for future investigations into aircraft survivability, weapons system design, and other areas where controlled experimentation and probabilistic modeling are integral to mission success.

6. FUTURE WORK

Although the current framework successfully demonstrates how Design of Experiments (DoE) and Bayesian network modeling can be integrated for first-order analysis, several promising avenues exist for refining and extending this approach. One key area involves the *dynamic modification* of the trade-space during experimental run-time, whereby intermediate results guide real-time adjustments to factor ranges or sampling densities. This adaptive strategy could exclude regions of negligible influence or, conversely, explore high-value areas in greater detail, ultimately improving computational efficiency and robustness.

Beyond adaptive experimentation, future investigations may also incorporate higher-fidelity models by introducing more detailed physics or sensor representations, such as terrain masking or advanced atmospheric propagation effects, to gain deeper insights into detection and survivability outcomes.⁵ Additionally, examining higher-order effects can further refine predictive performance; while this study focused on first-order interactions, real-world scenarios often involve second- or third-order dynamics that warrant systematic identification and modeling. As the dimensionality of the design space increases, scalable computing architectures, whether CPUs, GPUs, or distributed clusters, may become essential for maintaining tractable runtimes and enabling near real-time evaluation of experimental results. Finally, robust optimization and uncertainty quantification could

be achieved by integrating surrogate modeling, variance decomposition, or advanced Monte Carlo techniques, thereby guiding more informed decision-making under varied operational constraints.

By gradually introducing these enhancements, the proposed framework can evolve to address higher-complexity systems, bridging the gap between theoretical design objectives and the multifaceted realities of defense applications. In so doing, this work continues to lay the groundwork for principled, data-driven methodologies that adapt to evolving tactical requirements and emerging technologies.

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